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XAI for Industry 5.0—Concepts, Opportunities, Challenges, and Future Directions

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ABSTRACT Industry 5.0 has become a reality now and it is a paradigm that integrates contemporary innovations and concepts. Artificial Intelligence (AI) is a key component and asset of the industrial transformation which allows intelligent devices to perform functionalities such as self-examination, assessment, and evaluation autonomously. AI-based methodologies using ML and deep learning assist manufacturers and industrialists in forecasting service requirements and minimizing downtime. Recent research has discovered a remarkable change in the processes, systems, applications, and products in industries. Also, there is a significant challenge with the explainability of the decisions provided by the models using deep learning algorithms and their inadequate ability to be coupled with each other. Therefore, Explainable artificial intelligence (XAI) is required without compromising the efficiency of the models developed using deep learning algorithms. XAI investigates and develops algorithms, techniques, and models that produce human-comprehensible explanations of AI-based systems and can increase transparency and performance. The explainability nature of XAI will help humans understand the model and the reason behind the predictions, thus improving the model's transparency and the reliability of the outcomes. Furthermore, an Industry 5.0-enabled environment has a variety of data from varied sources, and this multi-source information must be fused to derive meaningful and optimal decisions. Therefore, all AI-integrated applications must derive actionable insights through information fusion. Hence, the adoption of XAI methodologies in Industry 5.0 can help humans make trustworthy decisions for critical applications requiring information fusion. In this paper, we present a state-of-the-art survey on adopting XAI in Industry 5.0. We discuss the adoption of XAI in various applications such as smart factories, smart Healthcare, E-Governance, smart transportation, Education 5.0, Agriculture 5.0, and Energy 5.0. Finally, some research issues and future directions of integrating XAI with Industry 5.0 are also discussed and highlighted to promote more study in the potential field.

INDEX TERMS XAI, AI, industry 5.0, smart factories, smart healthcare, E-governance, smart transportation, education 5.0, agriculture 5.0 and energy 5.0.

I. INTRODUCTION

THE ORIGIN of Industry 5.0 (I5.0) may be traced back to the manufacturing sector's movement from the era of Industrial Revolution to the age of digital transformation and beyond. Each successive stage highlights the industrial revolution's growth in the production process, which has altered the approach of working in the industry [1]. The Industrial Revolution has significantly transformed economies that were initially dependent on agriculture to the present generation of large-scale industries [2]. With consistent technological advancement, new production methods and techniques have come into existence, reducing cumbersome human efforts and marking significant enhancement in production, economy and overall societal standards. Although industrial manufacturing techniques have evolved throughout the years, the fundamental purpose of industry has remained the same to provide goods that improve the quality of life. Radical concepts have occasionally surfaced to alter the direction of industry throughout the history of manufacturing [3]. Since the inception of Industry 1.0 in the eighteenth century involving steam power, there have been giant leaps of progression to the present era with the application of information and communication technologies in various industrial processes. Network connectivity has enabled production systems involving complex computing technologies to be expanded to the development of digital twins in this generation. This has ensured communication across diversified facilities and helped acquire data, leading to production automation. The networking of all systems has created "cyber-physical production systems" and "smart factories," where people, components, and production systems interact over a network and produce almost autonomously. Industry 4.0 has revealed its ability to bring forth some fantastic advancements in manufacturing environments when these aforementioned enablers have come together [4]. For example, it has enabled machines to foresee problems and autonomously initiate repair procedures or self-organized logistics that respond to unforeseen changes in production. The production environment thus has become more digitalized, which has further opened up more flexible ways to deliver the appropriate information to the right person at the right time. Digital devices inside production units and in the field have ensured that maintenance professionals are equipped with relevant service history and documentation at the right time when they need to use them. This has also reduced wastage of time, providing scope for the maintenance professionals to invest their efforts exclusively on problem resolution [5].

Information fusion is an essential element in achieving the goals of I5.0. Information fusion involves integrating data from various sources, including IoT, sensors, third-party databases, and human inputs, to produce a more accurate, complete, and responsive view of the manufacturing process [6]. This integrated method for data processing aims to improve decision-making, increase production ability, and create a more responsive and adaptable manufacturing environment. Manufacturers can increase their operational intelligence by adopting information fusion [7]. For example, data from production floor sensors can be amalgamated with insights from quality control, supply chain logistics, and customer feedback to improve production processes, predict maintenance needs, and customize products to satisfy specific customer needs. This improves efficiency and productivity and enables greater customization and creativity, which meets I5.0's fundamental principles. Furthermore, information fusion enables humans and machines to communicate seamlessly, allowing real-time adjustments and collaborative problem-solving. This dynamic interaction promotes a more robust and flexible production system that can respond to dynamic consumer preferences and technological advancements [8].

The fundamental computing technologies of Industry 4.0 include AI, ML, Big Data, Cloud Computing, and cybersecurity, whereas Automation and Robotics, IoT, CPS, and AM constitute the physical part [9]. Together, these technologies realize the advantages of Industry 4.0 systems to enable agile, adaptable, and on-demand manufacturing, a crucial component of smart factories or smart manufacturing frameworks. Although implementing these technologies can help businesses gain a competitive edge and increase operational efficiency, there is widespread concern that doing so could result in the loss of jobs for low-skilled workers due to the high level of automation, which could lead to an imbalance in the economy and greater inequality in society [10]. Also, complete automation can generate false positive or false negative output, resulting in a waste of resources in the production process. Thus, despite the ability to digitally connect machines, enabling a seamless flow of data and optimization, I5.0 is considered to reinstate the human touch in the process to ensure sustainable manufacturing [11], [12]. Therefore, the need for collaboration between humans, robots, and digital technologies has been established.

Various studies have emphasised the inception, perception, applications, and challenges of I5.0. For example, the survey in [13] debated the various aspects of I4.0 and I5.0 and discussed various opportunities and challenges about their co-existence. The survey in [14] acts as a whistle-blower about the global transformation involving I5.0. Humans and their well-being are placed at the core of the manufacturing systems and the three most important characteristics of I5.0 namely human-centricity, sustainability and resiliency are discussed. The study presented a tri-dimensional perspective of I5.0 implementation: technical, reality, and application dimensions. In [15], the predominant enabling technologies and applications of I5.0 are highlighted, and further, a survey-based tutorial on the applications of I5.0 is presented. The study by [16] performed an exhaustive survey on published articles having the keyword "I5.0" and text mining was implemented to extract key terms and perform frequency analysis in this regard. The key enablers of I5.0 were thus identified, and various application areas, namely supply chain evaluation, enterprise innovation and digitization,

TABLE 1. Summary of literature survey.

Ref.No.	Contribution	Limitation
[13]	-The characteristics of Industry 4.0 and the transition to I5.0 are discussed. -The issues pertaining to the coexistence of Industry 4.0 and I5.0 are addressed	-The primary emphasis was the comparative analysis of Industry 4.0 and I5.0. -I5.0, and its associated challenges were briefly discussed without much emphasis on plausible means of improvement.
[14]	The human-centricity, sustainability, and resiliency aspects of I5.0 is discussed. -A three-dimensional architecture of implementing I5.0 is presented considering the technical dimension, reality dimension and application dimension	Societal issues in I5.0 implementation and challenges pertaining to the science and technology aspects are discussed without highlighting potential enablers to improve the same
[11]	-The complementation of I5.0 and co-evolution of I5.0 and Society 5.0 are explored. -The Goal, Value, Organization and Technological dimension pertaining to comparison of Industry 4.0 and Society 5.0 are highlighted establishing the dependency of one with the other.	The shared challenges associated with Human Cyber-Physical Systems, Greentelligent Manufacturing (GIM), Human-Robot Collaboration are discussed but the technologies to combat the challenges are not explored
[15]	-I5.0 concepts, the enabling technologies, and applications are discussed in detail. -Research challenges and open issues in I5.0 implementations are identified and highlighted	The generic challenges are enlisted and discussed superficially lagging in-depth analysis and exploration on potential remedial enablers for the same.
[16]	-Text mining tools are used to filter all publications where the "Industry 5.0" terminology is used. -The application of I5.0 in supply chain evaluation and optimization, enterprise innovation and digitization, smart manufacturing is discussed	The challenges pertaining to the various applications are not explored especially in the realm of AI, IoT and relevant enablers
This Paper	A comprehensive survey on Industry 5.0, its evolution stages and the relevant enabling technologies are discussed. I5.0 applications are primarily dependent on AI based technologies and the associated accuracy, transparency and interpretability issues are highlighted. The use of XAI to overcome such challenges is given prime importance in this paper, and its contributions to improving I5.0 applications are presented.	The proposed CAI and I5.0 integration's implementation constraints and benefits should be evaluated in further study.

smart manufacturing, and various other areas, were explored further. The study revealed the need for human-machine connectivity and co-existence as a significant theme for I5.0-based sustainable development. The various studies on I5.0 have highlighted its potential and applications involving key enabling technologies in association with human intervention, and associated challenges have also surfaced that are relevant to the transparency and traceability of the generated outcomes. In I5.0 handling the data from different sources necessitates using more transparent and understandable AI systems. XAI is essential for explaining AI-made decisions, supporting trustworthiness, and guaranteeing compliance. XAI elucidates the clear reasoning behind decisions made, assisting users in comprehending AI-powered recommendations that are vital for making safe, economic, and operational decisions. Additionally, XAI addresses inconsistencies in data and facilitates collaboration between humans and machines by ensuring optimal transparency and ease of use in AI operations. In addition, XAI guarantees that data fusion adheres to regulatory and ethical guidelines by offering complete audit trails, thereby promoting compliance and accountability in complex industrial situations. Table 1 depicts the Summary of Literature Survey.

The present survey thus emphasizes providing an exhaustive survey on the potential of XAI in I5.0 and relevant applications. XAI, often called Interpretable AI or Explainable Machine Learning (XML), is a type of AI in which humans can comprehend the justification for the decisions or predictions produced by the AI. It is a collection of procedures and techniques that make it possible for human users to understand and believe the output and results produced by machine learning (ML) algorithms. It thus helps to describe an AI model and understand the expected impact and potential biases. Also, it ensures the characterization of model accuracy and transparency of the outcomes involved in the AI-based decision-making process. Furthermore, I5.0, connecting numerous heterogeneous devices and applications coordinated with human input, may accumulate vast volumes of data through information fusion that could be used to build AI models. The interpretation of the model, its decisions, and various data biases must be understood for ease of use, safety, and a better understanding of these models' outcomes and data usage. Hence, XAI would suffice this need for I5.0's seamless adoption and successful collaboration of humans with machines. The use of XAI is exceptionally relevant in the context of I5.0-based applications, as building trust and confidence is crucial while putting AI-based models into production.

A. CONTRIBUTIONS OF THE PROPOSED SURVEY

The significant contributions of the survey are listed below:

- The proposed survey presents the research by providing various possible applications in which XAI can be deployed with in-depth analysis of XAI applications on service sectors like smart factories, healthcare, governance, transportation, education, agriculture, and energy. The survey is the first of its kind dealing with versatile applications in the depth of the problem domain
- Technical and non-technical requirements of the above-said applications are discussed with practical insight. This would foster the researchers to work on the key areas of various applications using XAI.
- The proposed survey discussed various challenges regarding the application of XAI, since it is a white-box model that may affect the data's privacy, integrity and security aspects and the point of contact with smart industry. This is addressed with the enhanced elucidation with the scope analysis of futuristic research

- The proposed survey discusses various use cases of XAI in smart industries, with detailed discussions regarding the societal contribution of XAI and I5.0.
- The proposed survey analyses the possibility of Man-Machine parallel living with the support of the XAI as the tool for the interaction and communication between the entities, with the enhanced capacity of elucidating to the end user about the perception of AI engines.

B. ORGANIZATION OF THE SURVEY

The proposed survey is organized with the following sections. Section II deals with the evolution of the XAI with the description of the types and processes associated with it. Section III discusses about various applications of the XAI in real-life demands such as Smart factory, Health care, Governance, Transportation, Education and Energy with the required discussion of use-cases and applications. Section IV provides various challenges in integrating I5.0 with XAI by discussing possible solutions and futuristic recommendations. Section V concludes with the motivation for the futuristic research with the Integration of I5.0 and XAI.

II. BACKGROUND

A. HISTORY OF INDUSTRY X.0

The industrial revolution disseminated the work from human hands to machines, eventually leading to the mass production of goods. In the late 17th century, people discovered the existence of abundant natural fossil fuels and innovative machines that drastically transformed human society's livelihood. In 1712, Thomas Newcomen invented the first atmospheric engine that harnessed the steam that is produced by boiling water to convert it to mechanical energy [17]. Initially, this design was used to pump out the water from the mines which significantly reduced human efforts. Later, in 1776, James Watt a Scottish inventor, improved the design of the steam engine [18] which paved a futuristic way for the concept of thermodynamics. This invention also has become the fundamental design that significantly broadens its usage and revolutionized it across other fields like manufacturing, steam turbines in power generation plants, locomotives, and steamships.

The first industrial revolution was named Industry 1.0 (I1.0), which started in Britain in 1784 and spread across the globe in the early 18th century. I1.0 utilized fossil fuels like coal, oil, and natural gas. When these fuels are burnt, they release enormous amounts of energy converted to mechanical energy. I1.0 was driven majorly by water-powered machines, which led to more incredible innovations, such as steamships, that changed the perception of ordinary people about time and space. Following I1.0 almost a century later, in the early 19th century, the world was stepping into the second industrial revolution, which was named Industry 2.0 (I2.0). This is also referred to as the "Technological Revolution" as many new technological innovations such as the internal combustion engine, effective methods of communication such as the telegraph and the telephone, railroad network, gas and water supply, chemical synthesis, and mass production in manufacturing and consumer goods, the first automobile and the first aeroplane were all invented. The I2.0 was between 1870 and 1914, which enabled the transportation of goods in large volumes from one place to another and paved the way for globalization.

After the I2.0, it was an era of computers. The first and second-generation computers that used vacuum tubes and transistors, respectively, came into existence during 1940-1960, and modern computers took their shape with the advent of time. The third industrial revolution, Industry 3.0 (I3.0), started slowly, and remarkable inventions happened after 1969 and were referred to as the "digital revolution". The third-generation computers that use integrated chip technology (ICT) were invented in the 1960s and leveraged fully automated machines to aid human operators or entirely replace humans. Later the fourth generation (1970) computers used microprocessor-based hardware technology. Industry 3.0 was a huge leap that allowed the use of Programmable Logic Controllers (PLC), robots, etc, to automate industry processes. It was during this period of transformation that more robots were employed, mainly in industries, to operate the tasks that were performed by humans. However, there was still human interference behind it.

The human participation in the automation processes in I3.0 was totally replaced with the up gradation of I3.0 to Industry 4.0 (I4.0) which was referred to as an era of Cyber-Physical Systems (CPS). In I3.0 the machines were only automatized but the data generated from them were to be carried forward by the humans for further decision makings and necessary actions were to be taken based on the decisions. CPS is basically the integration of hardware and software components that are connected together for seamless communication to perform the defined tasks. CPS is defined as the addition of physical systems like sensors and actuators, cyber systems like communication, computation, and storage, and protocols, on the Internet technology. Examples of CPS include intelligent robots that make a plan of action by themselves, intelligent buildings that control everything from lighting, and energy usage to user-specific functions, smart autonomous vehicles that drive themselves, and planes that fly themselves in controlled airspace [19].

I4.0 brought advances in intelligent machines, however, I4.0 has failed in cases such as the COVID-19, pandemic period when the entire world was at a standstill. Though industries have started adopting I4.0, by which the entire process is fully automated still the machines did not operate during the pandemic period. The greatest understanding that is derived from the COVID-19 crisis is that machines depend on human operators, human programmers, and human maintenance, and not everything could be automated. Human cognitive power and analytical capabilities are irreplaceable by machines. Similarly, the conventional methods adopted in the field of manufacturing in the era of I4.0 is very straightforward. The robots involved in the production phase are "dumb" as they would not consider the real users' choice but keep producing the parts in the pipeline and later would assemble it as a final product. This assembly system could not deliver the customization people wanted. The I4.0 framework mainly focussed on automation and thereby mass production was possible but customization was not possible. Later, the same business required the collaborative intelligence of both man and machine to enable customization. Therefore, I4.0 evolved into a new concept called I5.0, focusing on the interaction between man and machine.

I5.0 refers to human working with robots and smart devices. It adds the human touch to I4.0. During I4.0 the intention was to minimize human involvement and emphasis process automation. I5.0 is a new model that believes real progress lies in the collaboration between humans and machines. I5.0 does not replace I4.0, rather I5.0 is a compliment of I4.0. I5.0 framework utilises the features I4.0 and along with that adds the human operator to the framework. Hence interaction between human and machines are inevitable in I5.0 and focusses on personalization. I5.0 has the vision of an innovative, resilient, socio-centered, and competitive industry, which respects planetary boundaries and minimizes its negative environmental impact.

I5.0 is the enhanced version of I4.0 that promotes resilience, is human-centric and is sustainable. The human-centric approach promotes the worker's fundamental rights, the well-being of the worker, and a safe and inclusive environment. Due to the automation in I4.0, society might feel their jobs are taken away. As a result, industrial workers need to keep upskilling and re-skilling themselves for work-life balance [20]. I5.0 helps to improve the quality of life of the individual and society as a whole. Another essential aspect of I5.0 is the intelligent use of natural resources, which are reusable, re-purposed, and re-cycled, thereby producing sustainable products reducing waste, and improving the utilization of the available resources. The resilience of the current industry infrastructure is not as expected and has been proven unsuccessful during the recent pandemic period. The proposal and adoption of new industry policies should enhance the resilience of the industrial system during global emergencies or uncertainties in the production or supply-chain process.

The man-machine symbiosis makes remarkable imprints in the fields of healthcare, supply chain management, manufacturing, automobiles and etc. In manufacturing, for instance, Mercedes-Benz, to stand apart in the automobile business, invested in flexible manufacturing to offer personalization. AI-enabled collaborative robots (cobots) replace the robots in the plant. The workers in the manufacturing plant guide the cobot's arms to pick up and place heavy parts of machinery, which become an extension of the worker‘s body. The cobots can be reprogrammed with a tablet allowing them to handle different tasks depending on the changes in the workflow. Such agility has enabled Mercedes to offer personalized designs according to the user's choices. In this use case, the intelligence of the human and the ability of the robots are harnessed for improving the flexibility in design, personalization concerning the customer‘s choice, decision-making in critical situations with human intelligence, and speeding and scaling up the process with the help of the robots [1]. Similarly in healthcare, I5.0 can be used to develop new medical technologies and devices that allow the medical team to provide accurate and personalized care for individuals. For instance, essential human body vitals are collected and monitored remotely using wearable devices by medical practitioners. This allows the doctors or the nurses to intervene if the patient‘s data condition is alarming.

B. KEY ENABLING TECHNOLOGIES

The technology enablers of I5.0 include:

1) *Human-machine-interaction technologies:* I5.0 fosters a collaborative relationship between humans and machines, leveraging their complementary strengths for increased effectiveness. This is evident in the medical field, where I5.0 facilitates the development of inclusive tools catering to individual needs. Furthermore, technologies like AR/VR provide immersive training experiences, ensuring continued learning even during pandemics [21]. Additionally, cobots augment human capabilities in completing complex tasks, exemplified by personalized medical equipment that seamlessly adapts to a surgeon's hand.
2) *Bio-inspired technologies and smart materials:* Bio-inspired technologies that draw design inspiration from nature offer diverse solutions across various fields, including robotics, control systems, data analysis, and optimization. Bio-inspired robotics like gecko-grippers and virus-inspired nanosystems for drug delivery [22] demonstrate the adaptability of this approach. Notably, self-healing technologies inspired by skin regeneration hold the potential to minimize waste, extend product lifespans, and enhance safety when interacting with humans [23]. Meanwhile, self-healing materials inspired by spider silk promise longer-lasting, safer products for applications like prosthetics and wind turbine blades [24]. Furthermore, bio-inspired materials such as bio-adhesives and biologically made materials could be used for construction industries, 3D printing, and product development, promoting strengthened production. Also, bio-inspired AI algorithms, devised based on natural processes like Ant colony and swarm, could help develop more robust systems.
3) *Real-time based digital twins and simulation to model entire systems:* I5.0 is shaped by the digital twin in a number of ways, including predictive maintenance, which involves continuously monitoring the systems to foresee failures before they occur, process optimization, which simulates various workable methods as a process solution, quality control, which involves detecting defects early in the manufacturing process, supply chain management, which involves better inventory tracking, remote operations, which

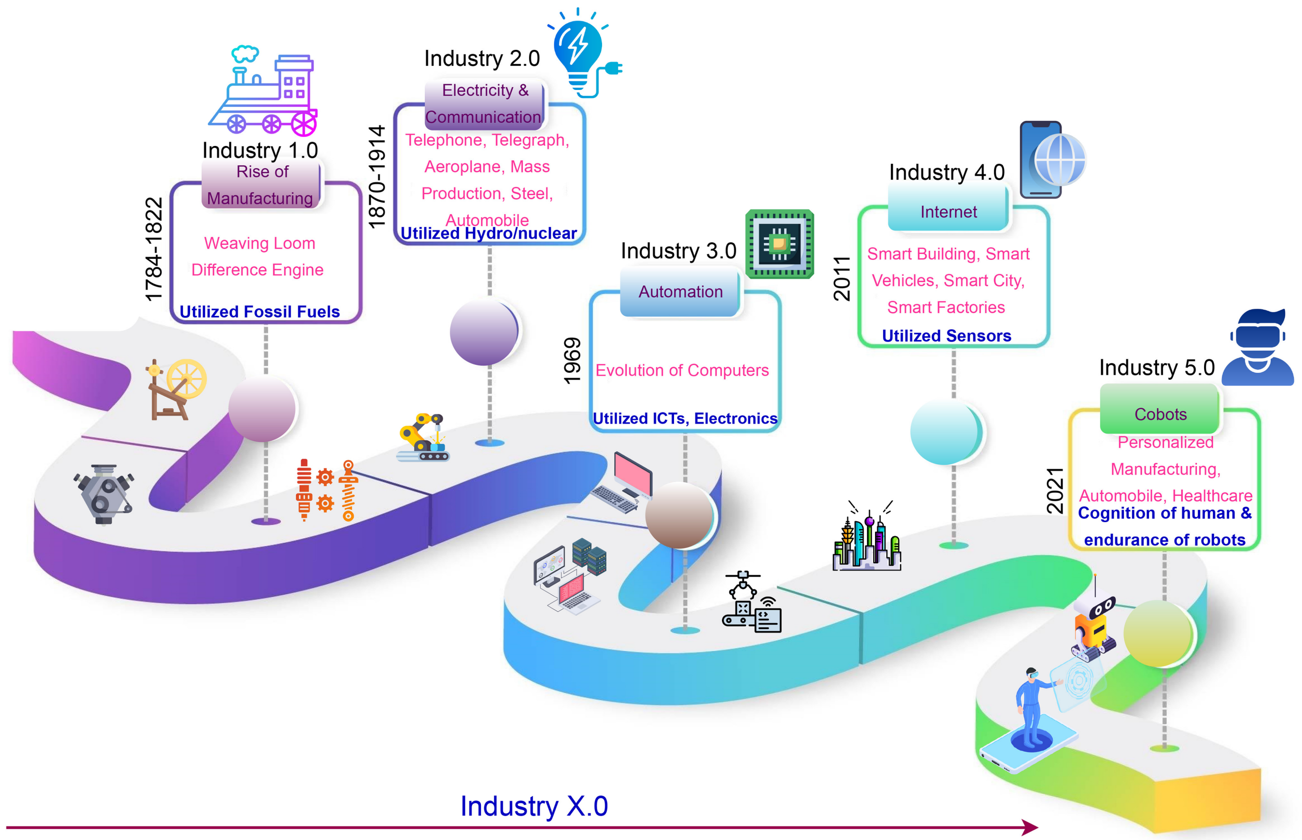

FIGURE 1. Evolution of Industry 5.0.

avoids human intervention in dangerous locations, training and simulation, which creates a realistic environment for improving trainees' skills, and customer experience, which involves allowing users to see and comment on the product [25]. To achieve human-centricity in smart manufacturing systems and move toward I5.0, the Human Digital Twin (HDT) idea is suggested as a crucial technique in [26]. HDTs are digital avatars of people to revolutionize the field of human-system integration by closely integrating human traits into the architecture and functionality of the system.

4) *Artificial Intelligence:* In line with the goal of I5.0 of fostering human-machine collaboration, the Brain-Computer Interface (BCI) acts as a bridge between the two in novel ways [27]. These include direct control over machines for complex tasks, real-time brain signal capture and analysis through cognitive augmentation [28], mental state monitoring for co-worker well-being, and enhanced rehabilitation through assistive BCI-based tools. Numerous sophisticated correlation analysis systems can adapt to novel or unforeseen circumstances. These encompass meta-learning, counterfactual reasoning, and transfer learning. The three primary components of I5.0 are people-oriented, environment-oriented, and resilience-oriented. To address power imbalances and strengthen grid resilience, the author in this study [29] suggested combining deep learning and transfer learning models to forecast power output in photovoltaic power plants.

5) *Technologies for storage, computing, and secure data transmission:* Blockchain, edge, and cloud computing technologies are all components of an ecosystem in the modern industrial revolution. Cloud computing offers resources on demand for applications requiring sophisticated data processing. Critical benefits including immediate decision-making, privacy in local data, and greater resilience are made possible in the edge computing. Blockchain facilitates transparent and safe transactions. The author of this paper [30] offers an empowered edge computing architecture that uses less energy and allows for real-time decision-making. The cloud acts as a central repository from which edge devices obtain insights. Real-time operations are performed by edge devices, which also send critical data to the cloud for further processing and preservation. Blockchain transfers data and transactions securely over the edge and cloud, ensuring transparency and confidence.

C. XAI

Building accurate AI models is the need of the hour, as it has applications in all domains ranging from healthcare to climate forecasting to autonomous vehicles to particle

TABLE 2. Privacy and security challenges in white-box XAI for smart industries.

Challenge	Overview	Potential Impact	Severity Level	Industries Most Affected
Transparency Trade-offs	Increased transparency can expose sensitive information about the system, posing IP and misuse risks.	Loss of competitive advantage, increased risk of reverse-engineering, and potential misuse of proprietary information.	High	Manufacturing, Aerospace, Electronics, Pharmaceuticals
Data Privacy Concerns	Detailed explanations may reveal private or sensitive customer/proprietary data.	Violation of data privacy regulations, loss of customer trust, and potential legal liabilities.	High	Healthcare, Finance, Retail, Telecommunications
Adversarial Attacks	White-box nature makes the system more vulnerable to attacks exploiting system details.	Disruption of critical operations, data breaches, and financial losses.	High	Critical Infrastructure, Energy, Transportation, Defense
Interpretability Limitations	Complexity still makes it challenging to fully explain the system's reasoning.	Difficulty in gaining trust and acceptance from end-users, and challenges in decision-making and oversight.	Moderate	All industries
Model Complexity	Complex algorithms for detailed explanations can impact performance and scalability.	Reduced system performance, increased maintenance costs, and limited deployment options.	Moderate	Manufacturing, Logistics, Energy, Utilities
Regulatory Compliance	Reconciling transparency with protecting sensitive data and trade secrets.	Regulatory fines, legal disputes, and reputational damage.	High	Healthcare, Finance, Telecommunications, Aerospace
User Trust and Acceptance	Detailed explanations may not align with user expectations and mental models.	Reduced user adoption, decreased system effectiveness, and challenges in user training and support.	Moderate	All industries

physics. The massive availability of computing resources and the increasing data sizes have geared the development of many complex non-linear models. The early phases of data modeling depended upon handcrafted rules and heuristics. The linear models and decision trees were then introduced to build models pertaining to several applications. However, the inception of ensemble and deep learning models have took over the entire stage by building efficient models. Meta-learning or the models that has the capability to create other models is the recent trend. The development of accurate AI models enabled a paradigm shift across various dimensions including expressiveness, versatility, adaptability and efficiency. The sophistication of the AI-enabled systems has grown to such an extent where no human intervention is even required for the designing and deployment of such systems. However, as the internal complexity of the models increases, the more opaque it becomes. The predictions made by the AI models can be non-intuitive thereby making it difficult for the people to understand what the model is actually doing. This would be a matter of concern when the decisions of the model are used for critical applications such as healthcare, military and so on. The capabilities of AI models can be further enhanced by allowing humans to closely collaborate with the machines so that solutions can be devised even for complex problems. Even though ML models are widely used, the "black box" nature of the model makes the human doubtful about the trustworthiness of the model when dealing with critical applications. Integrating the "explainability" feature with the AI system will enable humans to understand the model by explaining why a particular prediction is made, there by increasing the transparency of the model and trustworthiness of the results provided [31]. According to [32] "XAI will create a suite of ML techniques that enables human users to understand, appropriately trust, and effectively manage the emerging generation of artificially intelligent partners". The use of white-box models in XAI within smart industries causes privacy and security concerns due to their transparency. Table 2 shows the Privacy and Security Challenges in White-Box XAI for Smart Industries. The authors in [33] proposed the four key principles any XAI-based system should adhere to. This includes:

1) Explanation: The system should be able to provide valid evidence or reasons for the processes or outputs.
2) Meaningful: The intended user should be able to understand the explanations provided by the system.
3) Explanation Accuracy: The explanation should accurately reflect the processes of the system and/ or reasons for the predictions/ results provided.
4) Knowledge Limits: The XAI-based system should work only under the conditions for which it was designed and when it attains the required confidence in its result.

The significant motivating factors of XAI include trust, informativeness, accountability, causability, transferability, fair and ethical decision making and so on. This makes XAI useful for all the stakeholders including the users, the people who are affected by the decisions of the model and the developers [34]. An XAI model could personalize explanations based on the patient's medical history, explain the impact of lifestyle modifications like diet and exercise on blood sugar levels or impact of medicines. Adapting to new information obtained from the patient's body vitals', the model could provide tailored recommendations for adjusting medication or lifestyle habits. Such XAI model could be built on decentralized data without sharing sensitive patient information or could add noise to the data to protect individual privacy while preserving the overall utility of the model.

1) TAXONOMY OF XAI

XAI can be categorized into various types based on different dimensions [35]:

1) Ante-hoc vs. Post-hoc: This classification is based on the stage at which expainability is incorporated to the system. The ante-hoc model follows a new learning approach where the explainability is seeded into the model from the initial phase itself; whereas in the post-hoc approach, a black box phenomenon is followed

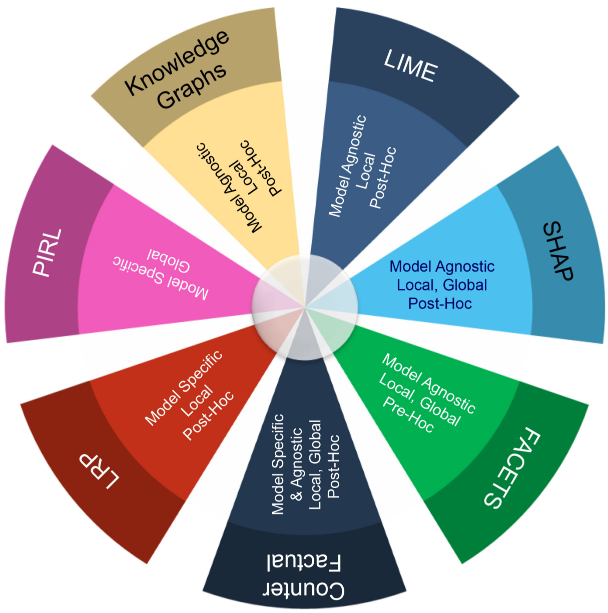

FIGURE 2. XAI Frameworks.

until the training phase is completed and explainability will be incorporated only during the testing time.

2) Model-Specific vs. Model-Agnostic: This classification is based on whether the XAI approach is dependent or independent of the ML model being utilized. It basically depends on the method's applicability to various models. The model-specific techniques are white box and they function by inspecting the internal structure of the model. Model-agnostic techniques in turn are black box approaches wherein they function by inspecting the relationship between input-output pairs of the models and does not depend on the model internals.
3) Local vs. Global: This classification is based on the scope of the explanation provided. Local explainability provides explanation about how a particular prediction is made for a specific input. This is significant in understanding what all unique features have contributed to the final decision at a particular instance. Global explainability, on the other hand, aids in deriving the features that are significant for the model output as a whole. It also helps in analyzing how a model learns by understanding the changes in the extend of a specific attribute employed in the decision making process of the model.

2) XAI FRAMEWORKS

XAI frameworks are tools that can be used for generating reports regarding how a particular model functions and provide explanations regarding its functioning. Fig. 2 depicts the XAI frameworks. The most popular XAI frameworks, LIME and SHAP are explained below.

1) LIME: LIME stands for Local Interpretable Model-agnostic Explanations. Each part of the name explains the features of this framework. This uses a model-agnostic approach and is a post-hoc model. Here, the scope of explanation is local. The predictions made using the LIME framework is individually comprehensible. That is, it provides explanations on the predictions made for individual data samples. It changes the individual data samples by altering the values of features and by analyzing the impact on the specific output. This framework aims at training local surrogate models. The authors in [36] have made an analysis on the advantages and the limitations of the LIME framework. The LIME provides a comparatively faster and simpler methods for locally explaining the black box models.
2) SHAP: SHAP is another XAI framework that is model agnostic, post-hoc and can provide both local as well as global explanations. SHAP stands for SHapley Additive ex Planations(SHAP). This framework can provide explanations for tabular, text and image datasets. Optimal credit allocations are connected with local explanations here with the help of Shapely values. It is derived from the shapley values of the game theory. Using this framework, each feature is assigned an importance value for a specific prediction [37]. One of the significant features of SHAP include the determination of a novel class of feature importance metrics. The authors in [38] proposed an interactive explainable framework using SHAP. This framework can be used for providing explanations about power system emergency control [39], intrusion detection [40], network anomaly detection [41] and so on.

Apart from the SHAP and LIME frameworks, various other algorithms also help in providing explanations about the predictions made by the model [42]. ELI5 is a python package that helps to examine the underlying ML model and explain the predictions provided. Similarly, Facets is an algorithm that is model-agnostic and ante-hoc. Here, the explanations can be provided both locally and globally. Fig. 2 shows the various XAI algorithms with their specific XAI taxonomy.

Even though explainability is not a mandatory feature for every domain, there are various domains where explainability is demanded. Explainability is significant in healthcare domain to help the medical professionals to understand the underlying reasons behind the diagnostic predictions provided by the system. Banking sector also require explanations to be provided to the user to understand why a particular credit was denied or to understand the reasons behind customer churn rate. Also, automobile industry will gain immense benefits by adding explainability along with its decisions. There are other domains such as agriculture, marketing, defense and a lot more that can benefit with the explainability that the AI provides. However, there are some challenges as well while deploying the XAI-based systems.

D. MOTIVATION BEHIND THE INTEGRATION OF XAI AND I5.0

The integration of XAI with I5.0 shifts the emphasis from mere automation to enhanced collaboration between humans and the AI systems. The key benefits that arise from the integration of XAI in the context of I5.0 are highlighted here. One of the primary advantages of XAI is transparency, with the help of which human operators can make informed choices based on a better understanding of the AI-generated recommendations. I5.0 focuses on improving collaboration between humans and automated systems. XAI facilitates the understanding of AI decisions by humans, enhancing the interaction between humans and machines. This allows humans to have control and make accurate choices based on the recommendations provided by AI. Furthermore, offering understandable explanations to the user helps in building trust between the humans and I5.0 systems. Trust plays a pivotal role in the human-AI collaboration, particularly in industries such as healthcare, finance, and autonomous systems. Humans can leverage the strengths of AI, thereby fostering problem solving and increased overall productivity. Accountability can be ensured for the systems, as XAI enables thorough audits. Explainability plays a crucial role in identifying and rectifying biases in the training data or algorithms. This, in turn, guarantees that AI applications strictly adhere to ethical standards. Furthermore, the integration of XAI aligns with regulatory requirements and standards. Safety and reliability are of utmost important in vital sectors such as manufacturing, healthcare, and transportation, while are important applications of I5.0. XAI guarantees the transparency and trustworthiness of AI system operations, which is essential in high-stakes environments where errors can have substantial implications. In conclusion, the benefits of incorporating XAI in I5.0 are manifold. From improving decision-making, building trust, promoting collaboration, and tackling ethical issues, XAI plays a pivotal role in shaping a future where cutting-edge technologies coexist harmoniously with human expertise.

III. APPLICATIONS OF XAI IN INDUSTRY 5.0

I5.0 aligns with the 9th and 11th sustainable development goals(SDG) proposed by the UN, namely Industry, Innovation and Infrastructure and Sustainable cities and Communities to promote a sustainable transformation [43]. This will promote a major reshaping of industrial processes with efficient information fusion. Therefore, this section presents the different applications of I5.0 with XAI. The inter dependencies between various I5.0 applications with XAI are drawn from a detailed analysis of (i) how the various cutting-edge technologies enhance the specific application, (ii) learning about the intricacies of data sharing, (iii) understanding workflows and (iv) aligning with the goals such as SDGs, and human-centric collaboration.

A. SMART FACTORY 5.0

Smart factories employ cutting-edge technologies like AI, Robotics and digital twins to lower costs, boost productivity, and enhance product quality [44]. Smart factories can make intelligent and self-organizing decisions in response to changes in their surroundings. Also, they can communicate with other manufacturers, suppliers, and clients to enhance collaboration and gain business intelligence. 3D printing, autonomous robotics, ML, predictive analytics, Blockchain, and cloud computing are some technologies employed in smart factory deployment. Using these technologies, the factory may increase productivity and efficiency by providing visibility in production. Also, intelligent factories can track and monitor their machine operations, enabling better maintenance and repairs [45]. Smart factories intended to make decisions about production scheduling, predictive maintenance, resource allocation, inventory control, and quality assurance. Decisions about data collection, analysis, and utilization must also be considered to improve processes and operations. Decisions should be made regarding integrating various technologies, including cloud computing, augmented reality (AR), virtual reality (VR), and the Internet of Things.

Decision models can forecast future results and trends, analyze data, and make choices on the most effective use of resources and employees. Predictive analytics, linear programming, decision trees, and neural networks are some of the decision models utilized in smart factories. These models can detect possible issues and provide solutions to increase productivity and lower expenses. Developing and supervising the factory's optimal control systems can also be done using decision models, guaranteeing that production is done efficiently and effectively [13]. XAI can be used in smart factories to improve decision-making and ensure that robot assessments are well-informed and consistent with predefined objectives. Multi-source information fusion in I5.0 help in integrating data from diverse sources such as machines, AI systems, sensors and humans, thereby providing informed, precise and timely decisions. Based on the context, data-level fusion, feature-level fusion, and decision-level methods could be utilized. Some of the commonly used models include JDL model, Bowman Df&Rm model, Luo-Kay model and Pau model [46]. Providing context-aware decisions help in maintenance scheduling, quality control, cost savings and improved sustainability. Robotic decisions can be explained using XAI so that people can comprehend why a particular action was taken. Moreover, XAI can locate and resolve production anomalies and enhance process monitoring and management. Finally, explanations that XAI can produce insights and suggestions for improving manufacturing processes and operations [47].

1) APPLICATIONS OF XAI IN SMART FACTORIES

Smart factories can employ XAI to increase operational effectiveness and decrease downtime. In the smart factory, XAI can assist in identifying and anticipating issues before they happen, reducing unplanned downtime and increasing

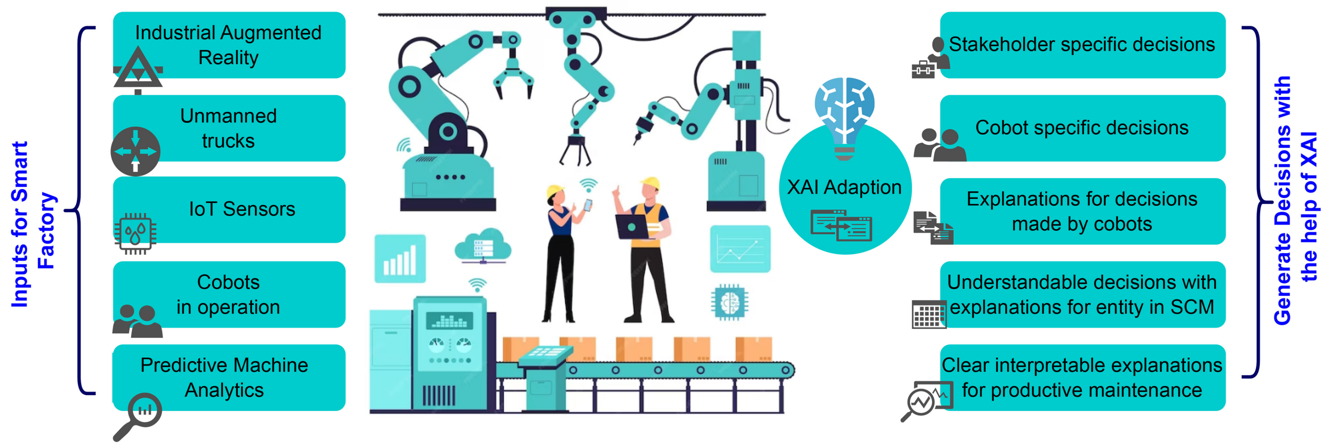

FIGURE 3. Benefits of XAI Adoption in Smart Factory.

output. Moreover, XAI can help to identify areas in the factory that require improvement to increase the effectiveness of the manufacturing process, enabling the production of high-quality goods in a short period.

Predictive maintenance can reduce industrial expenses by forecasting faults and extending component lifespan. Factories began monitoring their equipment recently, and most data gathered are in proper working conditions. Increasing intrusion detection accuracy, spotting unexpected failures, and adapting to changing operating and environmental conditions are significant obstacles to industry deployment. To combat this, AI-based predictive maintenance can be utilized in industrial scenarios where semi-supervised predictive maintenance is required to locate and address unforeseen issues. Fig. 3 depicts the various benefits of Smart Factory with the Adoption of XAI [48].

The significance of anomaly detection for quality prediction has gradually evolved due to increased data collection across various domains, including smart factories. Due to data imbalance in the anomaly detection model, numerous efforts have been made to increase discrimination efficiency in the current industrial process. Anomaly detection is crucial because it plays a vital role in predicting the quality of chromate processes, significantly impacting how complete they are. Additionally, collecting image data for monitoring during production is challenging, and prediction is problematic due to data imbalance. To provide a more intelligible assessment of faults with data from the chromate process, a model was developed using an unsupervised learning-based Generative Adversarial Networks (GAN) model, which performs training with only standard data images and adds a visual component to the Fast Unsupervised Anomaly Detection with GAN (F-AnoGAN) basis [49]. Thus, XAI has been applied in several applications of smart factories to make proper decisions and increase production. Although various techniques were adopted to accumulate information seamlessly, the challenges concerned with information fusion, like data quality for varied sources, should be foreseen. Consequently, liability upon integration and explanations generation for appropriate stake holders of I5.0 should not be overlooked.

B. HEALTHCARE 5.0

Various cutting-edge digital technologies, such as blockchain, IoT, cloud computing, and AI, enhance I5.0 in the Healthcare sector. Increased data accessibility, improved patient experience, and more individualized care are the benefits of these technologies. Digital health records, remote medical diagnosis and monitoring, virtual medical consultations, telemedicine, medical compressors, predictive healthcare analytics, and automated healthcare systems are some I5.0 applications in the healthcare sector. Healthcare providers can enhance the quality of care and minimize expenses by utilizing the possibilities of these technologies [15]. For the healthcare sector to successfully adopt new technologies and procedures and maintain its competitiveness and efficacy, decision-making is essential in I5.0. Decisions must be made regarding implementing the finest technology, policies, and infrastructure, the efficient use of data and the safety and security of patient data. One of the most crucial concerns with decision-making in healthcare is data sensitivity, and the decisions are also life surveillance determining factors. Therefore XAI explanations must be understandable to patients of different literacy levels and other entities at stake in the healthcare sector. Instead of one-size-fits-all explanations, user or machine-centric and dynamic explanations must be foreseen in further XAI models for the healthcare sector. Healthcare organizations must also decide the best approaches to control costs, enhance patient support, and ensure access to care. The healthcare sector must also decide how to effectively use the potential provided by digital healthcare, including how to best incorporate telemedicine and AI for medical support. To remain competitive and at the forefront of healthcare innovation, businesses must make wise decisions in the healthcare sector of Industry, 5.0 [50]. Doctors provide personalized recommendations and real-time information using XAI, which might help patients better. Additionally, by offering insights into the courses of therapy that are most effective for every patient and assisting in the identification of probable adverse events, XAI can be utilized to enhance patient safety. Patterns in patient records can be identified by employing XAI, which can then be used to improve public health initiatives. Finally, XAI can assist healthcare decision-making by providing helpful insights to medical professionals and healthcare systems [51]. Strategic technology investment like identifying the technologies that will have the most significant impact on patient outcomes and operational efficiency can be adapted to manage the costs. Introducing the technologies gradually can help the healthcare organisations to manage costs and risks. Telemedicine and AI can extend care to remote and underserved areas, reducing geographic barriers. AI-powered tools can improve diagnosis accuracy and treatment recommendations, leading to better patient outcomes. By automating routine tasks and optimizing resource allocation, advanced technologies can reduce wait times for appointments and procedures.

1) APPLICATIONS OF XAI IN SMART HEALTHCARE SECTOR

XAI can be successfully utilized in medical diagnostics and therapy, patient monitoring, medical decision support, drug discovery, etc. XAI can enhance illness diagnosis and therapy. Medical professionals can make more accurate and fast diagnoses with AI algorithms, which can spot useful trends in healthcare data that are difficult for humans to notice. AI can be utilized to provide individualized therapies and treatments for patients. XAI can be used to keep track of a patient's health and spot changes, and it can notify medical professionals when interventions are required. Doctors can receive real-time suggestions and guidance regarding patient care using XAI. AI algorithms can review patient data and deliver immediate feedback on the most appropriate action. XAI can be utilized to create more efficient clinical trial designs. Clinical trial outcomes can be predicted, trial designs can be made more effective, and suitable trial volunteers can be found using AI algorithms. Using XAI, drug discovery and development can be accelerated. AI algorithms examine vast amounts of data to pinpoint prospective therapeutic targets and forecast the efficacy of medication candidates [52].

Human life depends on clinical diagnosis models, so we must have the confidence to handle a patient as a black-box model directs. A study is done, and it provides examples drawn from the heart disease dataset and explains why explainability approaches must be chosen when utilizing deep learning methods in the medical field. The most common feature-based techniques, such as LIME and SHAP, have been discussed in the study. Although they serve relatively similar purposes, they are approached quite differently. These are crucial for exposing the opaque nature of black box models and ensuring their accountability. The methods have provided information about how the attributes lead to the prognosis. They have not provided any information regarding the features that help us to predict a particular patient's class. Integrated Gradients carry out this task. It is the only attribute that specifies the features that induce the model to provide a different forecast, called negative attributions, and the positive attributions assist in making a prediction. Thus, the most important lesson gained from studying these methods is that they all explain how different features contribute to the outputs of the model [53]. As the healthcare sector is sensitive, the data through information fusion should be dealt with carefully, focusing on data biases, time series information and interpretable explanations for the victims. Utilizing XAI with Healthcare 5.0 should not provide too much or too minimal information, as it may create unwanted chaos.

Reference [54] presented a survey emphasizing how well XAI fits into the healthcare 5.0 environment. The research developed a classification and segmentation architecture for medical images using XAI. The architecture integrates Convolutional Neural Network (CNN)-based Deep Learning methods with Federated Transfer Learning (FTL) for detecting the Covid-19 disease. Raw data gathered from COVID-19-affected patients are efficiently normalized by the developed FTL-assisted CNN autoencoder classifier, which divides the dataset into five classes. The system evaluates the classifier's prediction and provides the decision using the proposed explainability module and the explainable diagnostic module. Hence, XAI can be used in various healthcare system applications to provide better patient support and assist medical practitioners. XAI can significantly contribute to public health initiatives by providing insights into complex patterns within patient records. By understanding these patterns, healthcare organizations can identify risk factors, develop targeted interventions, and improve overall outcomes.

C. GOVERNANCE 5.0

I5.0, the next Industrial Revolution, collaborates humans with machines, enabling advanced manufacturing and production capabilities by integrating digital, physical, and biological technologies. As such, it is essential to ensure that efforts to build and deploy new technologies are made responsibly and with a consideration of the implications for global governance. Good governance in I5.0 is essential to ensure that these new technologies are used for the benefit of society, not for the help of a few [13] and assure improved business agility. To overcome this situation, the government and other private organizations should join hands to craft various policies and required regulations that enforce public protection and nurture liable innovation. This includes promoting data privacy and security, ensuring the ethical use of AI, and developing guidelines for using smart manufacturing technologies. In addition, various governments and industrial organizations should focus on creating a universal framework that summarizes the values of noble governance in the industry practices [44]. The integration of XAI with I5.0

poses various challenges, especially in the context of smart manufacturing. However, these challenges could influence several future research directions in this domain. One of the significant areas is that of human-centric interactions [55]. It becomes mandatory to design intuitive and context-aware frameworks and interfaces. Similarly, evaluation metrics for evaluating the trustworthiness of a system is of utmost importance. Developing light weight and real-time XAI systems which are scalable is another critical area of research. Sustainability is one of the major focus areas of I5.0. Hence, green manufacturing could be studied for future research.

Furthermore, for efficacious governance, decision-making with appropriate and adequate user-specific explanations for various decisions made in I5.0 applications is mandated. As such, decision-making needs to be done in a more informed and data-driven manner. This requires advanced analytics to make informed decisions and automated decision-making systems to help with real-time decision-making. Additionally, decision-makers need to be able to access and analyze data to make informed decisions promptly and quickly. Ultimately, decision-making in I5.0 requires an agile approach in a rapidly changing environment like Governance and policies must be curated through careful examination and evaluation.

1) APPLICATIONS OF XAI IN GOVERNANCE IN INDUSTRY 5.0

Various frameworks for realizing the ethical principles were suggested in the literature, such as responsible data design, a process-based framework for governance, and actionable principles [56]. Furthermore, these rich manufacturing use cases and further adoption of this I5.0 in various sectors require standards for building open components compatible with all other parts for effective integration. Furthermore, it can provide exploratory reports on monitoring business integrity. Although various reference architectures for Industry 4.0 were available to address the technology adoption and usage concerns, one baseline architecture is required for I5.0 [57]. One such architecture for Industry 4.0 is evaluated by proposing a new reference architecture in detail to promote its usage for I5.0 in bringing humans in the loop for secure AI in smart manufacturing [58].

Most of the issues concerning the usage of sensitive data in AI have been alleviated by XAI adoption as it provides insights into the black box of AI. It also helped satisfy the data privacy policy enforced by the European Union‘s General Data Protection Act [59], [60]. XAI can help to improve the governance of AI systems by providing greater transparency and understanding of how decisions are made and why. XAI can help ensure that decisions made by AI systems are logical, fair, unbiased, and based on accurate and up-to-date data [61]. XAI can also be used to help identify potential areas of risk or areas of improvement that can be addressed to improve the governance of AI systems. XAI can also evaluate the AI system‘s performance and identify areas for additional training or other interventions. XAI can provide comprehensive justifications for predictions and guarantee that people comprehend the decisions made by AI algorithms.

D. TRANSPORTATION 5.0

Smart transportation in I5.0 focuses on creating autonomous technologies and systems to improve the effectiveness of transportation networks, including advances in intelligent traffic systems, automated driving, and digital tools to monitor and manage traffic flow. Autonomous vehicles are anticipated to become more prevalent, using cutting-edge sensors and AI to enable them to drive safely and effectively. To improve traffic planning, real-time data gathering and analysis will be made possible by cutting-edge technology, such as 5G networks. Blockchain-based solutions will also assist in protecting data and guarantee its accuracy. By improving routes and easing congestion, smart transportation systems will also significantly reduce emissions [62].

Features of smart transportation systems are

1) Autonomous Vehicles: In I5.0, autonomous vehicles are predicted to transform the transportation sector completely. Advanced technologies like ML, computer vision, and AI will empower autonomous vehicles [63]. Autonomous vehicles can avoid obstructions and other traffic while spotting patterns and making decisions based on the information they gather [64].
2) Smart Traffic Control: I5.0 is anticipated to introduce smart traffic control technologies. These systems identify traffic patterns and modify signal timings based on those patterns using sensors, cameras, and AI algorithms. This will aid in reducing congestion in traffic and improving transportation efficiency [65]. To secure IoT data in smart traffic control systems, measures include encryption, authentication, regular updates, network segmentation, and intrusion detection systems. Blockchain enhances data integrity through its decentralized structure, immutability, transparency, consensus mechanisms, and smart contracts. Blockchain architecture prevents data tampering, ensures transparent transaction verification, and automates processes through self-executing contracts.
3) Connected Infrastructure: I5.0 will heavily rely on connected infrastructure. It will enable the Internet of Things to be connected to transportation networks, providing real-time information about traffic congestion and other circumstances. As a result, transportation authorities can take the initiative to boost the effectiveness of transportation networks [66].
4) Smart Ticketing: In I5.0, smart ticketing systems will increase transportation productivity. Passengers can buy tickets online and pay for transportation using their mobile devices. As a result, fewer physical tickets will be required, and buying tickets will be significantly simpler overall [67].

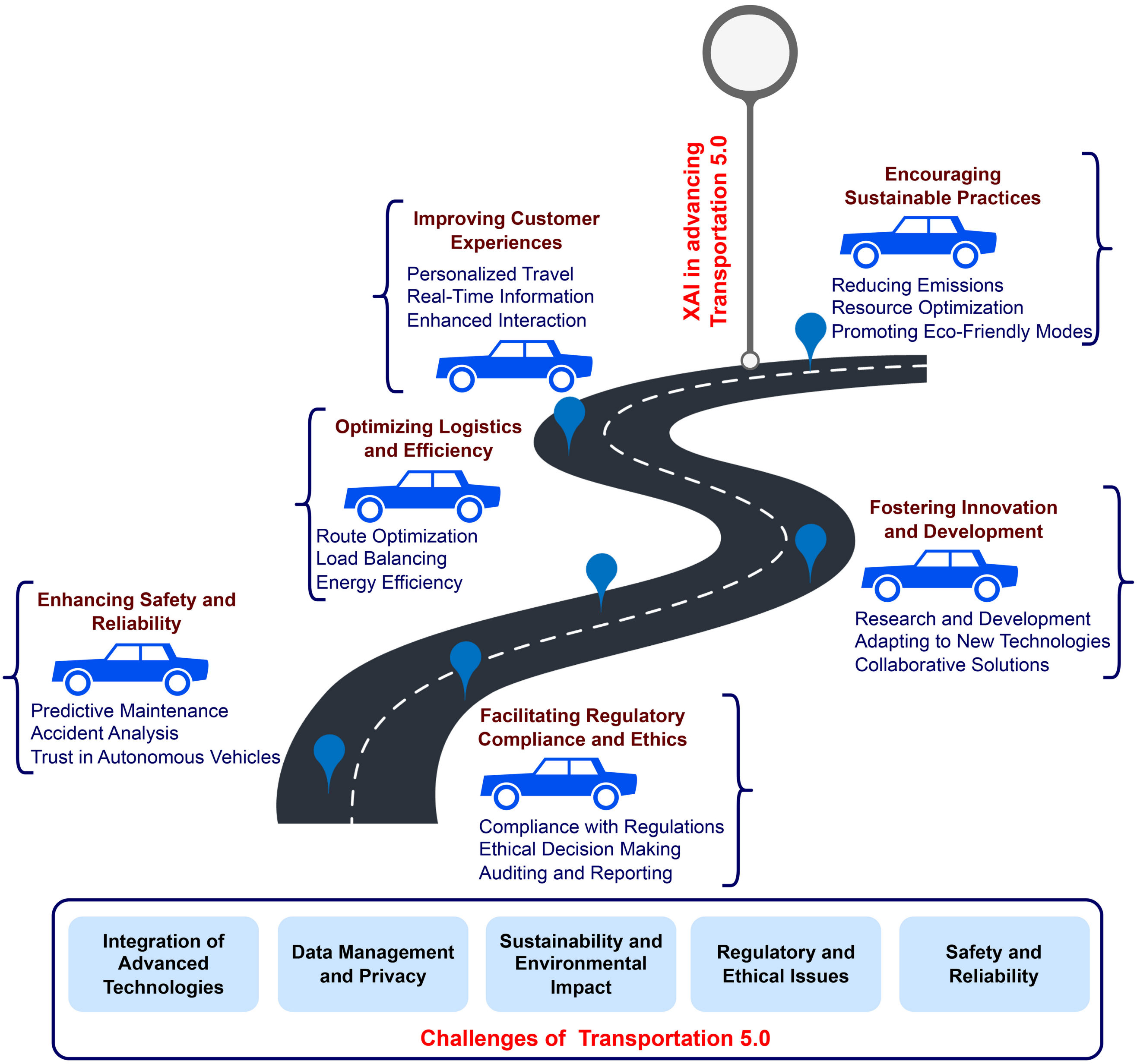

FIGURE 4. Challenges in Transportation 5.0 and XAI in advancing Transportation 5.0.

5) Sustainable mobility: I5.0 is anticipated to focus significantly on sustainable mobility. This will include employing electric vehicles to reduce the carbon impact of transportation and employing renewable energy sources like solar and wind to power automobiles [68]. Various challenges of Transportation 5.0 and benefits of XAI in uplifting or mitigating those challenges in transportation 5.0 are depicted in Fig. 4.

Smart transportation in I5.0 is mainly about integrating advanced technologies such as AR, Big data, IoT, and AI into the transportation system to improve efficiency and decrease costs. The decision-making in this domain will involve integrating these technologies with the existing transportation systems to optimize operations. Decisions like load optimization, path selection, predictive maintenance, traffic flow optimization, and others fall under this category. The decisions will be generated from data analysis, AI algorithms, and data gathered from multiple sensors. The decisions adopted should minimize transportation costs, increase system effectiveness, diminish the adverse impacts of traffic on the environment, and guarantee the well-being of the passengers [11].

Smart transportation requires decision-making to optimize the flow of people and goods. This involves making choices on deploying resources like trains, buses, and ride-sharing services, as well as positioning traffic lights and road signs. AI systems that use information from cameras, sensors, and other sources to discover patterns and foresee probable traffic disturbances are frequently used to make these judgments. The choices taken by these algorithms can assist in increasing safety, decreasing congestion, and improving traffic flow [62].

1) APPLICATIONS OF XAI IN SMART TRANSPORTATION SYSTEM

There are numerous ways that XAI can be applied to enhance the smart transportation system. Better traffic flow and vehicle routing forecasts can be made with XAI, resulting in more effective transportation systems. XAI can also generate more precise predictions of travel duration and delays to improve traffic management and safety. Additionally, XAI can more accurately detect and forecast traffic accidents and other occurrences, enhancing safety and easing congestion. Better visualization tools like 3D maps and AR, which can aid drivers and passengers in better understanding their surroundings and making better judgments, can also be provided by XAI [69].

XAI improves the transparency of decision-making in autonomous vehicles by providing detailed explanations of how and why decisions are made, particularly in crucial circumstances. The transparency fosters public confidence, guarantees adherence to regulatory norms, and enables ongoing enhancement of driving algorithms. During an unavoidable crash scenario, XAI explains the justification behind the vehicle's selection of a particular path. XAI plays a crucial role in providing auditable decision trails, which are necessary for legal and insurance reasons. This facilitates effective communication with all parties involved [70].

Utilizing a multimodal deep learning architecture, traffic flow prediction employing XAI can assist in predicting traffic patterns, allowing transportation planners to create plans to ease congestion and boost safety [71]. Route Optimization using XAI can optimize routes for different transportation modes, such as public transit, cars, and buses, to reduce travel time and improve efficiency [72]. XAI for Ridesharing can match passengers with drivers and optimize routes to reduce wait times and improve the overall passenger experience [73]. Public Transport Management using XAI can manage public transportation fleets, helping reduce delays and improve efficiency [74].

E. EDUCATION 5.0

Data-driven, personalized, and hands-on training are the primary objectives of smart education in I5.0. It will integrate established instructional strategies with cutting-edge technologies like VR, AI, AR, and ML. It will be used to acquire the abilities and information required to perform in the digital sector. Smart Education would employ AI to offer customized learning paths, ML to provide individualized feedback, AR/VR to deliver immersive learning experiences, and smart analytics to monitor the progress.

Education 5.0 is an idea of digital transformation in teaching and learning that uses the most recent innovations and market trends to produce a more individualized, engaging, and successful educational experience. Big data analytics, AI-powered learning systems, VR, and AR are all examples of technology that could cater for the learner's needs. Education 5.0 in I5.0 improves learning and teaching environments by reducing the space between industry and academics [75].

AI in education can significantly change how students learn in the classroom. A customized learning plan is developed for each student based on their interests and needs when deploying AI to personalize education. AI can also automate the grading process, freeing teachers' time to focus on instruction and student interaction. AI can also be utilized to design individualized learning environments and virtual classrooms. In addition, AI can assist with data analysis and research, enabling teachers to make better-informed choices about how to teach their kids. Teachers can give pupils a more interesting and productive learning experience with the aid of AI [76].

Various issues in Education 5.0 incorporation, advantages of XAI and its benefits in Education 5.0 is depicted in Fig. 5.

1) APPLICATIONS OF XAI IN EDUCATION 5.0

The application of XAI in education can completely transform the way students learn. XAI improves personalized education by analyzing the unique learning styles, strengths of individuals. It customizes content by comprehending students' requirements through their interactions, adapting learning paths, and modifying content hardness in real-time. XAI presents explicit feedback and suggestions while identifying motivational factors to enhance user participation. This facilitates the establishment of a flexible and adaptable learning environment and enhances transparency in the educational process, enabling students and educators to better understand the explanations behind customized modifications and choices.

XAI can assist in locating student learning gaps and offering solutions to fill them. Additionally, instructors can utilize XAI to determine the most effective ways to teach specific ideas and better understand their pupils' strengths and limitations. Additionally, by measuring learning outcomes with XAI, educators may better understand what works and what doesn't in the classroom [77]. XAI helps analyse student performance data to precisely identify specific points where the student is struggling. Knowledge derived from XAI aids tutors in adjusting their approach to individual needs based on the advancement of each learner. XAI recommend applicable educational resources or methods applied to other similar students that have shown success. XAI continuously analyses how students respond to different techniques and learning materials. XAI improves communication between teachers and students by providing appropriate, correct, and precise evidence-based justifications for academic decisions and educational improvements.

Learning analytic methodologies for student assessment can be automated using XAI. To accurately analyze student

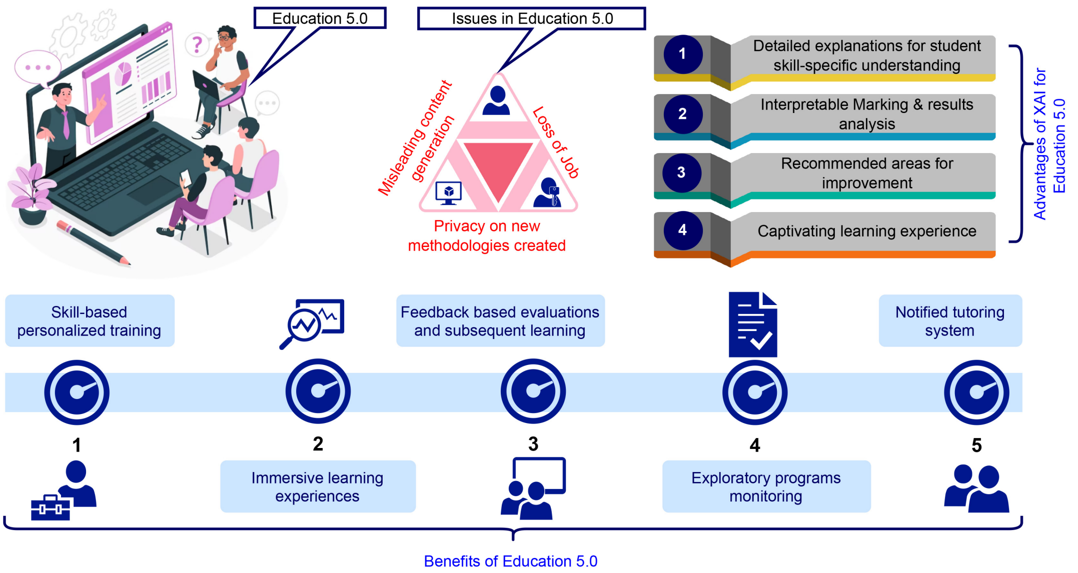

FIGURE 5. Issues in Education 5.0 and leveraging the XAI for better learning.

performance on tests and other assessments, AI-based algorithms are deployed [78]. Research on intelligent tutoring systems (ITS) focuses on creating instructional strategies to replicate students' fundamental needs, circumstances, and talents, such as subject expertise, meta-cognitive abilities, and mental states, and provide individualized instruction. Students can benefit from personalized learning opportunities made possible by XAI. Based on each student's unique requirements and preferences, learning experiences can be tailored using AI-based algorithms [79]. Each student's requirements can be satisfied by creating courses with XAI. Course content customized to each student's requirements and interests can be created using AI-based algorithms. Additionally, they are employed to offer specialized student support. Students can receive individualized support from XAI through timely reminders and notifications, customized resources, and feedback. Students are given interesting content, such as interactive learning modules, games, and simulations, utilizing AI-based algorithms [80]. Inappropriate information generation, more curated contents, provoking data should be monitored while tailoring XAI applications for Education 5.0.

F. AGRICULTURE 5.0

"Agriculture 5.0" refers to applying cutting-edge technologies to enhance agricultural output and boost farming productivity. It uses sensors, robots, drones, big data, and IoT to increase agriculture yields and animal husbandry. The goal behind the agricultural revolution known as "Agriculture 5.0" is to use cutting-edge technologies, including automation, robotics, IoT, AI, and big data, to increase the productivity and sustainability of agricultural processes. This revolution aims to make global food production more effective and sustainable. Improvements in agricultural inputs, precision agriculture technologies and procedures, and the incorporation of agricultural data into decision-making processes are all part of it. Agriculture 5.0 aims to bring about a more efficient and sustainable agricultural revolution that will also increase employment and improve people's quality of life [81]. AI in agriculture 5.0 applies AI tools like computer vision, big data, and ML to enhance crop production and farm operations. AI can help farmers manage resources more effectively, automate labor-intensive operations, and enhance agricultural yield [82].

Issues in agriculture

1) Crop selection: The identification of suitable crops to be cultivated depends on regional circumstances, market trends, and consumer preferences.
2) Irrigation Management: Choosing when and how much water to apply to crops and determining how much crop productivity would be affected by water scarcity.
3) Fertiliser application: Choosing the right fertilizers, how much to apply, and how to minimize the environmental impact.
4) Weed control: Identifying the appropriate weeds to remove and figuring out how to deal with them, including applying herbicides, crop rotation, and other cultural practices.
5) Pest Management: Choosing the best pest control procedures, including the use of pesticides and other pest control techniques, that cause the least amount of disruption [83], [84], [85].

XAI can be utilized to develop decision models to manage the issues and improve crop production. Leveraging the power of XAI in agriculture 5.0 and its benefits are depicted in Fig. 6.

1) APPLICATIONS OF XAI IN AGRICULTURE 5.0

Several research works have been done by applying ML algorithms and AI techniques for decision-making, such as prediction and classification in the agriculture domain.

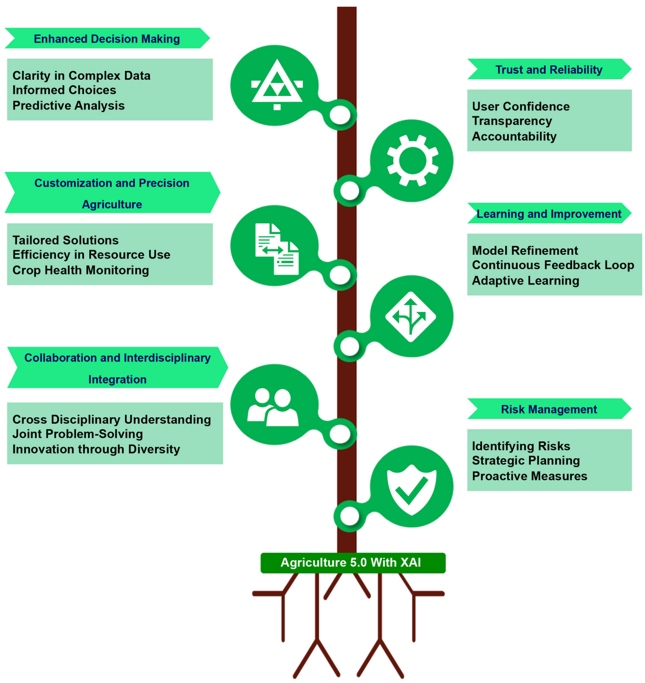

FIGURE 6. XAI Adoption in Agriculture 5.0.

In those works, it is unclear how the learning process is done and the rationale behind the decision-making. To solve this issue, an XAI has been applied to the agriculture dataset, including soil, weather and other parameters pertaining to crop yield for data analysis. The data analysis provided the decision that no-tillage management can enhance the yield of the maize crop. The XAI and interpretable ML algorithms have provided the answers, such as the useful parameters important for prediction, the association between input and response parameters and the reason behind the predicted value for a particular observation. Further, it also provided information on whether other ML algorithms have provided the same answer. The XAI has proved that the questions were unanswered by other ML algorithms but XAI and interpretable ML can enhance the explainabiity and trust in AI [86]. The explainability nature has been explored by applying XAI to the dataset of wheat head detection. The dataset has information collected from various repositories related to different growth stages and lighting conditions. It is challenging to analyze such data, leading to poor model stereotypes. To solve this issue, a pre-trained YOLO model was applied to detect wheat heads using the SHAP library. In the developed model, SHAP helps to detect a partially overlapped wheat head. The model proved to have the ability to minimize the cost and time in the supervision of wheat crops [87]. An XAI framework with ontology design and knowledge map model was proposed for digital agriculture to provide human-understandable explanations for the decision and results. The framework was developed to provide algorithms, definitions and information for developing data mining-based models. Further, Agriculture also proposed an agriculture computing ontology for explaining the knowledge mined on various datasets [88]. A system was developed to detect plant disease using XAI techniques. In the model, the XAI method, namely Gradient weighted class activation mapping++, was applied to identify plant disease and emphasise the regions covering the leaves that lead to classification [89]. Storage, security, reliable and accurate results are some of the factors to be considered in the evaluation of XAI for Agriculture 5.0.

G. ENERGY 5.0

In I5.0, "Smart Grid" refers to integrating cutting-edge technology into the electricity grid to improve its effectiveness and efficiency. This includes adopting smart meters, sensors, and distributed energy resources (DERs) to facilitate a two-way flow of electricity from utilities to end consumers. In addition to using analytics, automation, AI, and IoT, the smart grid in I5.0 also uses these technologies to improve resource efficiency and control the distribution and consumption of electricity. The ultimate objective of the Smart Grid in I5.0 is to lower prices, boost dependability, and aid in the reduction of greenhouse gas emissions. The larger power consumers of today are industrial sectors who started to use the SG even decades ago before knowing the term as a whole. The major concerns with today‘s energy systems are greenhouse gas emissions, the economics involved in rapid hikes in electricity prices, worsened grid reliability, and weakened energy security (When electricity is transmitted via electric vehicles or other plug-in vehicles) apart from the traditional electric grid‘s issues such as lack of storage capabilities, voltage level distribution, interrupted power service, and internal equipment failure with complex wiring. The consumer‘s role is consuming and producing electricity by sharing the energy when required, thereby termed a prosumer. The concept of Energy 5.0, which encompasses the various industrial interactions with energy development [90], enforces human intelligence in the modeling and analysing smart energy systems.

Though the smart grid offers effective solutions for power management by moving operations and decision-making to the source of power generation, some of the major issues are:

1) *Cybersecurity:* As grid systems become increasingly interconnected and automated, the potential for malicious cyber activity increases.
2) *Data privacy:* As more and more data is collected from smart meters and other sources, it‘s important to ensure that data is collected, stored, and used securely. Data privacy of sensitive information about data usage, and sharing is also a major concern.
3) *Network reliability:* With so many connected devices, it‘s important to ensure that the grid remains operational and resilient in the face of physical or cyber threats.
4) *Cost:* While many technologies that make up the smart grid are cost-effective, the upfront costs of deploying

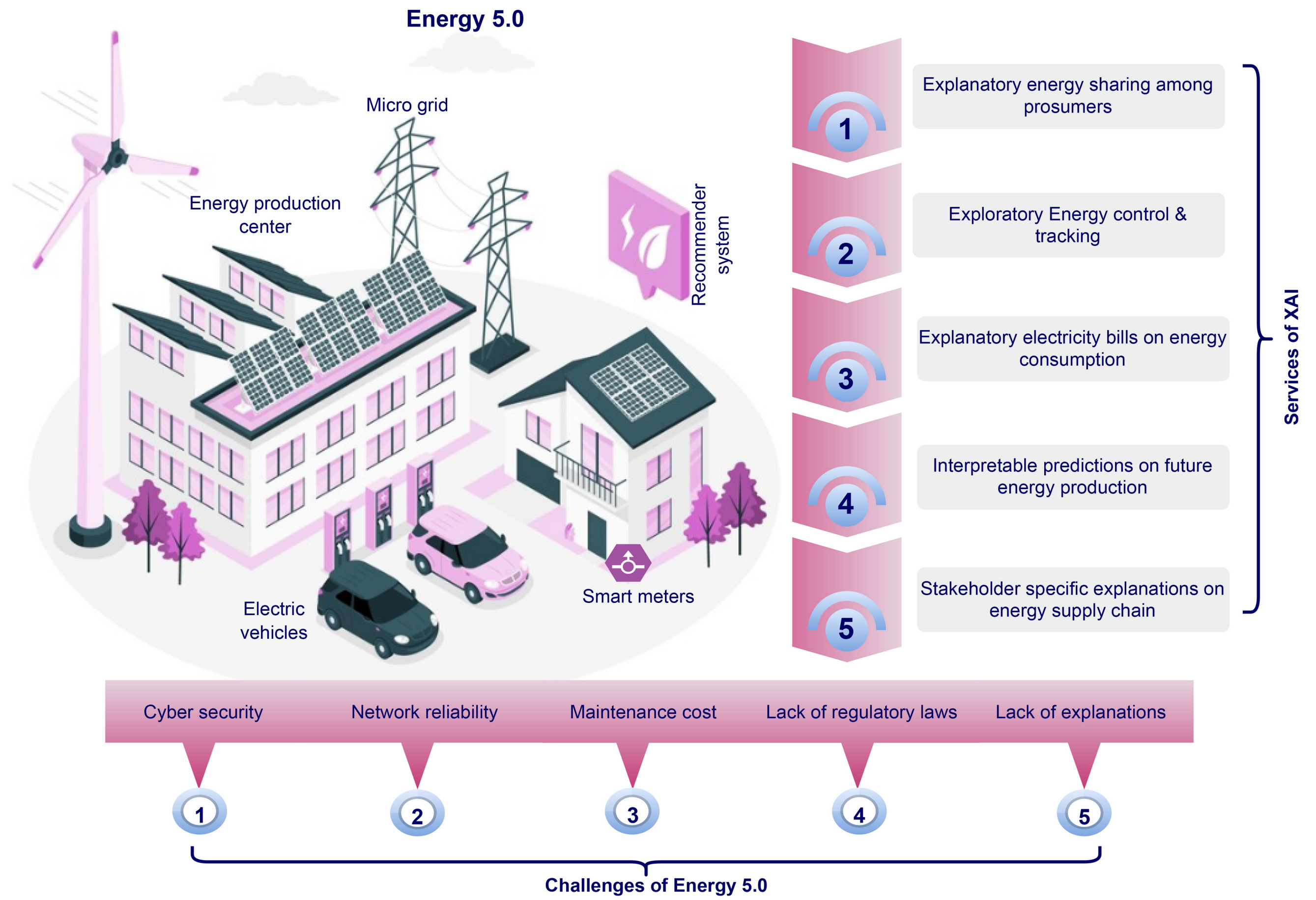

FIGURE 7. Issues in Energy 5.0 and XAI Services in Energy 5.0.

the infrastructure can be high and require long-term investments.

5) *Regulation:* SG technologies are complex and, in some cases, may require new regulatory frameworks in order to ensure their safe and effective use.

XAI is required in smart grids because it allows decision-makers to better understand the system's decisions. Decision-makers can use XAI's practical insights to assist them in making decisions that are consistent with their goals and objectives. With the assistance of XAI, decision-makers will be able to address potential issues and obstacles in the smart grid system more quickly. By offering more insight into the underlying data and operational processes, XAI in smart grids can enhance decision-making processes. Engineers and operational staff can benefit from XAI by making faster, more accurate decisions that improve efficiency and reduce costs [91]. Smart grids employ XAI to defend decisions made by AI-enabled technologies. Users must comprehend the reasoning behind the system's judgments if we hope to increase user confidence in these systems. Additionally, XAI can assist operators in spotting abnormal behavior and modify the system as needed. Additionally, XAI can offer perceptions into how the system operates and assist operators in anticipating potential issues before they become serious. Finally, XAI can enable automated systems that can react swiftly to modifications in system behaviour to maintain systems' smooth operation and minimize hazards. Challenges in Energy 5.0 and services of XAI in Energy 5.0 are depicted in Fig. 7.

1) APPLICATIONS OF XAI IN ENERGY 5.0

XAI contributes to numerous Energy 5.0 solutions. Anomaly detection is one of them, where XAI may be used to find abnormalities in the smart grid and notify operators if anything is not working properly. This can help reduce downtime and improve system reliability. XAI for predictive maintenance can also predict when certain smart grid components need maintenance or replacement. This can help reduce costs and improve the efficiency of the system [92]. XAI in Load Forecasting can be used to forecast the demand for electricity and help optimize the use of resources [93]. Demand Response using XAI can accurately predict the electricity demand and help optimize resource use of resources [94]. This can help reduce costs and improve the system's efficiency. Furthermore, XAI for Security and Privacy can detect potential threats and vulnerabilities in the smart grid. It has been used for intrusion detection in the smart grids [95] with various algorithms and their usage for intrusion detection. This can help reduce the risk of malicious attacks and improve the security and privacy of the system.

H. SUMMARY OF APPLICATIONS

The different verticals of I5.0 were discussed with the application of XAI. These include smart Factors, healthcare, governance, transportation, education, agriculture 5.0, and energy 5.0. I5.0 in all these verticals was discussed with the employability of XAI in generating user-specific explanations by considering different entities at stake. Utilizing XAI models in these verticals will enhance the performance and expose some open issues, namely, security and privacy, interoperability, transparency, and lack of regulatory concerns in adoption. XAI's models can explain the decisions based on different users' needs and perspectives. This, in turn, may help ensure trust in adopting AI-based I5.0 applications to the fullest.

IV. OPEN ISSUES, CHALLENGES AND FUTURE DIRECTIONS

XAI is projected to play a significant part in I5.0, such as manufacturing, logistics, and healthcare, combining AI, IoT, and automation. Although many research works have been proposed for XAI, the intricate and black box features of XAI raise some open issues in I5.0-related applications. Table 3, explains the open issues and challenges face the integration of XAI within I5.0. Addressing these challenges through the proposed future directions significantly enhance the impact and reliability of AI systems in industrial applications.

1) *Explainability-by-Design:* The applicability of XAI is presently available for existing models. The role of incorporating explainability principles into a model is to ensure enhanced transparency from the beginning while retaining the model's performance. Integrating the explainability elements into the intrinsic AI training and development process can provide more straightforward explanations regarding the model performance.
2) *Context-aware explanations:* Explanations about a specific industry sector is essential to highlight the unique intricacies associated with it. It helps in dealing with essential customization so that the dynamics of human-machine collaboration, safety standards, and operational objectives are improved. The effectiveness of Human-machine collaboration also benefits from the customized explanations pertaining to individuals as they can highlight the explainability factors related to specific user preferences and their responsibilities.
3) *Trust-worthy explanations for complex models:* XAI also helps improve the models' trustworthiness. Currently, deep learning models are widely used for I5.0 applications. Although their responses are quite opaque, still integrating XAI could help in improving the trustworthiness, simplicity, and accuracy by analyzing the explanations.
4) *Real-time explainability:* Industrial applications are not only memory-constrained due to the extensive use of edge devices but also time-constrained as the operations need to be performed in real-time. Therefore, XAI methods need to be designed so that they are computed in real-time to comply with the timely decision process.
5) *Integration with diverse data formats:* I5.0 considers many diverse data types that can be obtained from sensors, cameras, or any other device. This results in multimodal data types such as images, text, audio, and other information formats. Specialized XAI methods must be created so that the data types can be integrated and explained within the same framework accordingly.
6) *Human-centered interaction and trust:* The integration of human-machine collaboration within the industrial environment is based on interfaces through which humans communicate to the machines and the trust that the machines would do the job appropriately. Both tasks, i.e., interfaces and trust, can be improved by analyzing the XAI explanations. However, XAI techniques should be designed to cater for both the interfaces and trust while explaining the model.
7) *Standardization and Evaluation:* XAI is an emerging field and so does I5.0. There is a lack of standardization and evaluation metrics in the domain of XAI and I5.0. Therefore, there is a significant demand for the development of standards and metrics that would provide a guideline for using such emerging technologies and a standard way to validate them. The explanations provided by the XAI models are evaluated using two different ways: subjective and objective. In the case of the subjective metric, domain experts' input and the random audience input are used for evaluation. In terms of objective metrics, metrics are devised specific to the task, such as contrafactual and example-based methods and, on the other hand, based on models like model trustworthiness and attribution. Some commonly used objective metrics are Simplicity, sensitivity, completeness, stability, computational cost, monotonicity, correctness of explanation, confidence of explanation, fidelity, generalizability and sparsity. Some subjective metrics are comprehensibility, satisfaction, efficiency, pervasiveness, trust, effectiveness, justifiability, interestingness, goodness and transparency. However, there is no proper standard evidence for the implementation of these metrics [96].
8) *Ethical considerations and regulations:* Security and privacy of individual and industrial information is one of the utmost priorities for human-machine collaboration to work. XAI explains, however, that these explanations can be compromised and manipulated through various attacks. Some of the potential risks or ethical concerns that might arise with wider adoption of XAI in Industry 5.0 applications include bias, misinterpretation of explanations, accountability, and

excessive reliance on AI [97]. In this regard, XAI methods should be designed to comply with ethical concerns and data privacy rules.

9) *Transparency:* In I5.0, some XAI systems may give transparency, but not all the knowledge is included to comprehend a decision or prediction that might make an unconvinced system for users. For the transparency issue in the XAI, Ehsan et al. [98] introduced Social Transparency (ST), a viewpoint on XAI that considers socio-organizational context. The authors interviewed AI users and practitioners to investigate ST's technological, decision-making, and organizational consequences and implications. This survey expands the design area for Human-Centered XAI techniques. Some XAI systems may deliver simplistic explanations that cannot represent decision-making complexity and undermine system trust. Based on this new problem, Tubella et al. [99] proposed a mechanism for evaluating AI moral limits based on inputs and outcomes. Provable rules that control system inputs and outputs are presented with clear moral principles. This technique increases explainability by allowing stakeholders to review the system's interpretation and use of moral principles. Focusing on inputs and outcomes helps evaluate AI systems and examine their morality. This strategy improves AI system users' confidence and well-being by clarifying its moral standards. In some XAI systems, the underlying data required to train the model may be private or inaccessible. This situation makes it hard to evaluate the model's correctness or comprehend data biases. Winter and Carusi [100] examined the limited acceptability of AI and ML technologies in hospitals due to a lack of epistemic openness. The authors discussed how an AI application for the early identification of a rare respiratory ailment was developed and how tight collaboration between AI engineers, physicians, and biological experts might improve transparency. Overall, transparency is a significant issue in the XAI-related I5.0 applications. In the future, more appropriate countermeasures should strengthen openness to enable comprehension and confidence while still being practical and efficient for XAI.

10) *Interpretability:* Even with substantial openness, XAI may not be sufficient to comprehend an AI model fully. This can make it difficult for consumers to trust the model's conclusions and for regulators to determine if it is compliant. XAI depends on humans' capacity to comprehend and explain the behavior of an AI system. A standardized definition of interpretability makes comparing and evaluating XAI solutions impossible. As AI models become increasingly integrated into decision-making processes, consumers and stakeholders must be able to rely on the model's output. Interpretability enables users to comprehend how the model arrived at its conclusions, fostering trust and ensuring that the AI system's decisions can be held accountable. In [101], the authors stated transparency and accountability in AI systems would help people understand AI algorithm decisions. In particular, black-box AI and algorithmic decision-making are increasingly challenged. Their study analyzes how AI explainability affects user trust and attitudes toward AI. It conceptualizes causability as an antecedent of explainability and a vital cue of an algorithm. It analyzes how they affect the user-perceived performance of AI-driven services concerning trust. Moreover, interpretable models allow developers to easily find and fix AI system biases and faults, improving prediction accuracy. Deep learning models are used in application like temporal multilabel classification (TMLC) models, which assign multiple labels to input sequences, like video activity recognition and annotation. Debugging TMLC models is challenging due to the exponential growth of label combinations and the need to understand how errors propagate within sequences. Researchers have developed DETOXER [102], an explainable, interactive visual analytics tool for debugging video activity recognition models to address these challenges. DETOXER provides multiple scopes of explanations and outputs, including frame-level outputs, video-level explanations, and global-level explanations. This helps users detect errors of various scopes and types through exploration and interactive visualization. In certain fields, including healthcare and finance, regulations require explanations for AI-driven decisions. The ability to interpret aids in meeting these requirements and ensuring ethical compliance. The authors in [103] announced that AI systems must comply with human rights, the law, and the public welfare. To maintain public safety, social stability, and innovation, governments and politicians must develop real policies within moral, legal, and cultural frameworks. Well-established tools and design techniques can assist governance in reducing negative episodes, developing trust, and stabilizing society. Regulations that encourage socio-legal and technical compliance can boost innovation. The policy must assign human accountability for AI system development and deployment to fill decision-making gaps caused by automation. Regulating AI helps promote well-being for everybody in a sustainable world. XAI can help create ethical and law-abiding AI systems that benefit healthcare and finance. XAI in policy-making ensures AI systems align with human values, encourages trust, and promotes openness, creating a more responsible and beneficial AI ecosystem. Future directions for the interpretability of XAI will focus on achieving a balance between interpretability and performance in AI models and developing techniques that adapt to user requirements, scale efficiently, and

address domain-specific challenges. This will enhance the comprehension, credibility, and adoption of AI across various applications.

11) *Security:* Other than transparency and interoperability, security is another issue that should be considered to protect sensitive data and prevent AI system. Secure XAI solutions protect privacy, user trust, and AI-driven judgments while providing explanations. Security also reduces the possibility of adversarial assaults on AI models that could mislead and hurt. Users' comprehension and trust in ML models is crucial to their successful use in cybersecurity. Transparency and explainability increase as black-box ML algorithms make critical predictions. Cybersecurity professionals require explanations for ML model outcomes. Recent research has focused on explainability approaches, white-box interpreter assaults, and explanation characteristics and metrics. Security characteristics, cybersecurity threat models, and black-box assaults on explainable models are missing. Kuppa and Le-Khac [104] presented an XAI taxonomy for cybersecurity-related security features and threat models. A new black-box attack analyses gradient-based XAI algorithms' consistency, accuracy, and confidence security. The developed system misleads the classifier and explanation report on three security-relevant datasets and models without impacting the classifier output. These insights can improve XAI security and robustness. Moreover, explainable approaches boost the deployment of ML methods, which combat growing cybersecurity risks. Counterfactual explanations are popular because they assist users in comprehending black-box model decisions and identifying mutually exclusive data examples that would influence outcomes. Recent XAI research has focused on explainability methodologies, white-box interpreter assaults, and explanation characteristics and metrics. Model explanations can add attack surfaces to underlying systems, although research is few. Membership inference, model extraction, poisoning, and backdoor attacks can compromise system privacy using explanations. This research examines counterfactual cybersecurity features and threat models to fill the gap. Based on the above analysis, Kuppa and Le-Khac [105] suggested a black-box XAI attack to violate classifier confidentiality and privacy. The approach is evaluated with cybersecurity domain datasets and models, showing its effectiveness under real-world threat models and emphasizing XAI security. Nguyen and Choo [106] developed a conceptual framework called "Human-in-the-Loop XAI-Enabled Vulnerability Detection, Investigation, and Mitigation" (HXAI-VDIM) is proposed. HXAI-VDIM integrates security analysts or forensic investigators into the man-machine loop, leveraging XAI to combine AI and Intelligence Assistant (IA) to amplify human Intelligence for proactive and reactive processes. HXAI-VDIM aims to create an interactive, iterative loop between humans and machines using security visualization, enabling human Intelligence to guide the XAI-enabled system and generate refined solutions, emphasizing the importance of security in XAI. Future directions for XAI security will emphasize augmenting robustness, privacy, and dependability while preserving explainability. This includes developing secure explanation methods, addressing potential attack surfaces, implementing privacy-preserving techniques, and developing human-machine integrated frameworks to foster user trust and safeguard sensitive data. Quantum computing could further enhance XAI's model interpretability, which would excel in extracting relevant features from complex, fused data and can easily interpret data of different dimensionality more effectively. Further, it could assist in generating counterfactual explanations through quantum mechanics [107]. Similarly, emerging technologies like brain-inspired computing or Neuromorphic systems could play a vital role in I5.0 with XAI [108]. Some notable benefits are reduced latency through faster computation at the edge, anomaly detection and energy efficiency. Though these technologies are individually beneficial, integration would result in enormous challenges like scalability and lack of standardization.

V. CONCLUSION

AI has faced adoption challenges due to its lack of transparency and reliance on diverse data. Also, information fusion is vital for AI-based applications as it learns through the assorted data collected from varied sources. The trust in AI systems was significantly regained with the evolution of XAI with its explanations for AI's decisions. Information fusion is required for XAI to derive optimal decisions and appropriate explanations to intended users. As I5.0 and its applications rely on AI, XAI for I5.0 would significantly enhance the performance of the Industrial Revolution and its trust in adoption. Therefore, this survey presents the concept of utilizing XAI for I5.0, the need for information fusion, viable applications, and challenges in adoption and future research directions. The survey first provides insights into the need for XAI for I5.0 and briefs the major contributions to the research. Followed by the historical background of the Industrial Revolution, the taxonomy of XAI, its framework, and the motivation behind the amalgamation. Consequently, metaverse applications in different verticals of I5.0, namely Smart Factories, Healthcare, Governance, Transportation, Education 5.0, Agriculture 5.0, and Energy 5.0, have various implementation issues. Finally, the open issues and challenges, namely, Transparency, interoperability, and security, were identified, with feasible solutions for resolving those issues. Furthermore, insights for future research directions were also discussed.

TABLE 3. Open issues, challenges, and future directions for XAI in industry 5.0.

Open Issue	Description	Challenges	Future Directions	Impact	Importance
Explainability-by-Design	Incorporating explainability principles from the beginning of the AI model development to enhance transparency and performance.	Balancing complexity and transparency, ensuring explainability without sacrificing model performance.	Develop frameworks and tools that integrate explainability at all stages of AI development.	Enhances trust and user understanding of AI systems.	High
Context-aware Explanations	Developing explanations that are specific to industry sectors to improve human-machine collaboration and operational objectives.	Adapting explanations to various industrial contexts and user roles while maintaining accuracy and relevance.	Design adaptive systems that tailor explanations based on the sector-specific needs and user roles.	Increases effectiveness of AI in specialized sectors.	High
Trustworthy Explanations for Complex Models	Enhancing the trust, simplicity, and accuracy of deep learning models through XAI, making their operations less opaque.	Developing methods to effectively simplify complex model outputs into understandable explanations.	Focus on methods that provide deeper insights into complex model behaviors and decision-making processes.	Critical for adopting AI in sensitive applications.	Very High
Real-time Explainability	Designing XAI methods that can deliver explanations in real-time to align with the quick decision-making needed in industrial applications.	Creating fast and efficient explanation processes that do not compromise the timeliness of decision-making.	Develop lightweight, efficient algorithms capable of operating under strict time constraints.	Essential for time-sensitive operations like manufacturing.	Very High
Integration with Diverse Data Formats	Creating specialized XAI methods that can handle and explain multimodal data from various industrial sources.	Managing and synthesizing information from heterogeneous data sources to provide coherent and actionable explanations.	Innovate cross-modal explanation techniques that seamlessly integrate different data types.	Supports better decision-making from complex data sources.	High
Human-centered Interaction and Trust	Improving interfaces and trust through XAI explanations, ensuring that human-machine collaboration is effective and reliable.	Designing intuitive interfaces that facilitate understanding and trust between users and AI systems.	Enhance interface designs and develop trust-building features within XAI.	Improves usability and acceptance of automated systems.	High
Standardization and Evaluation	Developing standards and metrics for evaluating and implementing XAI in I5.0 to guide technology usage and validation.	Establishing universally accepted metrics and standards in a rapidly evolving field.	Establish international standards for XAI applications and evaluation methodologies.	Facilitates consistency and comparability across systems.	Very High
Ethical Considerations and Regulations	Ensuring XAI methods comply with ethical concerns and data privacy regulations to maintain security and privacy.	Navigating varying international data privacy laws and ethical standards.	Create ethical guidelines and compliance checks within XAI systems.	Protects user data and adheres to legal standards.	Very High
Transparency	Addressing the need for comprehensive transparency in XAI systems to foster user confidence and comprehension of AI decisions.	Overcoming the complexity of AI models to provide understandable and useful explanations.	Advance methods that offer more detailed insights into AI operations and decision-making.	Builds trust and enables better oversight of AI systems.	High
Interpretability	Ensuring XAI models are interpretable so users can understand how decisions are made, which is crucial for trust and regulatory compliance.	Achieving high levels of interpretability without impacting the performance of complex AI systems.	Focus on developing techniques that offer clear, understandable explanations suited to non-expert users.	Ensures AI decisions are justifiable and accountable.	Very High
Security	Focusing on securing XAI solutions to protect sensitive data and prevent AI system abuse, addressing threats like adversarial attacks and ensuring robust explanations.	Protecting against adversarial attacks and ensuring the integrity of explanations in the presence of potential security vulnerabilities.	Strengthen defenses against attacks on AI models and develop secure protocols for explanation generation.	Enhances the safety and reliability of AI applications.	Very High

Future research would focus on developing an XAI framework for providing user-specific explanations in different application scenarios of I5.0 by utilizing Human-Computer Interaction and Natural Language Processing tools for multi-lingual different user-level explanations. Furthermore, the safety of these systems can be enhanced by adopting federated learning and can be efficient and fastened with less resource utilization through transfer learning models.

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